

LFEWS 2.0 - Local Flood Early Warning System

User Manual

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1. Introduction

Welcome to the LFEWS 2.0 (Local Flood Early Warning System) User Manual. LFEWS is a comprehensive platform designed to provide real-time updates and historical data regarding flood monitoring, weather observations, tidal behaviors, and critical mappings for evacuation and hazard risk reduction.

2. Getting Started

To access the system, navigate to the application URL provided by your administrator. You will need a registered account with the appropriate role (e.g., Administrator, Reviewer) to access secured modules.

- **Login:** Use your assigned credentials (email and password) to log in securely.
- **Roles:** Depending on your user role, you might have restricted access to certain management or settings panels.

3. Dashboard

The **Dashboard** is the main overview screen presented after logging in. It features:

- **Summary Visualizations:** High-level representations of current system status.
- **Budget Visualizations:** amCharts displaying MOOE and Capital Outlay allocations for the current operational year.
- **Quick Controls:** Buttons to manually pull or refresh recent data for Water and Weather parameters instantly.

4. Monitoring System

The core operation of LFEWS relies on active data collection endpoints seamlessly integrated into the application.

Water Level Sensors

- **Overview:** View all connected water level sensors deployed across the designated river systems.
- **Management:** Administrators can Create, Edit, or Delete sensors.
- **Data Pull:** Authorized personnel can manually trigger data pulls to reflect real-time river conditions.

Weather Stations

- **Overview:** Lists all weather monitoring equipment synced with the platform.
- **Management:** You can add new stations, update current details, or remove obsolete configurations.
- **Data Pull:** Similar to water sensors, users can initiate manual synchronization for latest weather reading metrics like precipitation, temperature, and wind speed.

Lunar Tides

- **Tide Graphing:** Visual representation of "Tidal Extremes" utilizing the WorldTides API integration. Keep track of incoming high and low tide phases relative to the coastline.
- **Synchronization:** Sync the tide projections within the system for updated graphing calculations.

5. Maps and Geolocation

LFEWS emphasizes visualized locational data for better response coordination during calamity scenarios.

Locator Map

A consolidated view of all available devices and stations pinged onto an interactive map area.

- Click on any specific sensor or station marker to view a **Device Status Popover**, reflecting its current online visibility and its latest recorded reading limit.

Hazard & Flood Risk Maps

Provides multi-layered map configurations to outline areas that are highly susceptible to flood hazards and other topographical risks.

- You can manage recorded risk areas within the "Hazard Maps" and "Flood Risks" modules.

Evacuation Centers

An accessible view charting all identified Evacuation Centers in the region, aiding in strategic disaster response deployment.

6. Reports

The **Reports** module serves as the primary tool for archiving and exporting metric histories.

- **Water Level Data:** Retrieve historical flood depths corresponding to specific timeframes or sensor nodes.
- **Weather Observation Data:** Pull out past weather conditions recorded over time for accurate forecasting references.

7. Administration & Settings

Reserved primarily for system administrators, these menus dictate the core structure of LFEWS 2.0.

System Settings

- **Categories:** Configure core metadata like Appropriation Categories.
- **Budget Config:** Manage active "Budget Years" and designated "Fund Sources".
- **Rivers Modules:** Manage predefined target rivers aligned with sensor installations.

Users & Roles Configuration

- Navigate to the **Users Management** screen to add, revoke, or edit user profiles.
- **Roles Manager:** Directly create and customize User Roles defining exact permissions across the platform interface. Users can be selectively switched between respective access levels easily.

System Document generated for LFEWS 2.0